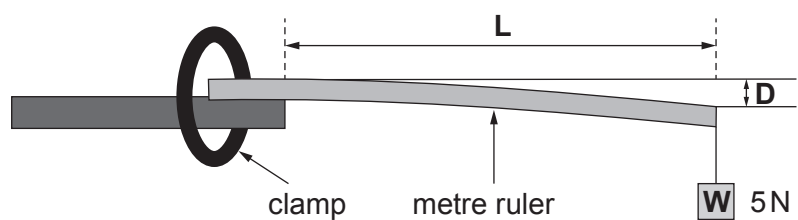


Answer **all** questions.

1. Students are investigating how the stiffness of a wooden metre ruler depends on its length. They use the apparatus shown below. The weight, **W**, on the end of the ruler remains constant at 5 N.

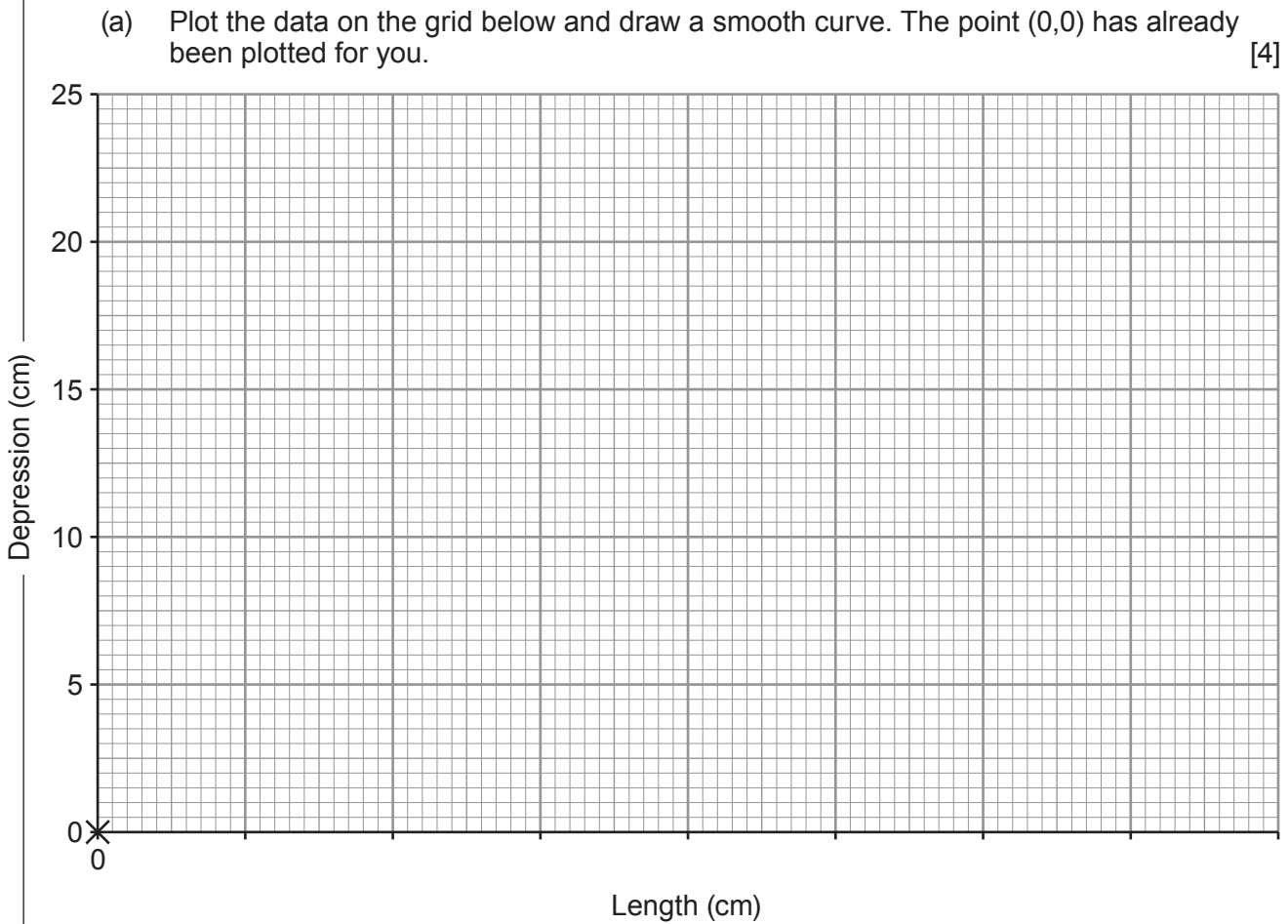


The length, **L**, is varied. For each length, **L**, the depression, **D**, of the end of the ruler is measured.

Their results are shown in the table below.

Length L (cm)	Depression D (cm)
0	0.0
30	1.0
40	3.0
50	6.0
60	10.0
70	15.0
80	22.0





- (b) It was expected that depression, **D**, would be directly proportional to length, **L**. Explain whether the graph shows a proportional relationship. [2]

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- (c) Use information in the table and the equation:

$$\text{stiffness} = \frac{\text{weight}}{\text{depression}}$$

to calculate the stiffness of the ruler at a length of 50 cm. [3]

stiffness = N/cm

